Group 3, “$20”
Kiser
e4USA engineering
2 December 2024

Part A

This cart handle project addresses a rather simple problem: make a physical solution to the discomfort mr. Kiser experiences while pushing the short double-decker lab carts (similar cart pictured below, courtesy of The Home Depot).

The top of the cart’s handles are 33.5 in high, which doesn’t do any favors to the 6’3” Mr. Kiser, especially given the 2 slipped discs in the lower back. Said medical issue results in pain while twisting and bending down being inadvisable. In a separate strain, we believe the handle should ideally work for people of any height, the specific heights for which we chose to be between 4’6” and 6’6”, since anyone under 4’8” will be able to comfortably use the supplied handle of the cart (based

off of our minimum measured joint angles while pushing a loaded cart) and anyone over 6’6” is impractical to cater to, given the rarity of such height (only 0.5% of men 30-39 yrs old at or over 6’6”, according to the 2007-2008 NHANES) and the amount of extra material that would be required.

In spite of the fact that we are focusing on Mr. Kiser right now, This handle could potentially help many people, because if the handle we create can be adjusted to fit multiple carts of the same style, then any tall person already in possession of a similar cart could utilize this cheap handle to make the cart easier and less uncomfortable to maneuver. Whether this would come into practice every day or hardly ever, just about any tall person would be glad to have it, since any time you’re moving something heavy or unwieldy, these carts will be your, the primary stakeholders, saving grace. As for secondary stakeholders, we will be focusing on moving/shipping companies’ upper management and students. The companies often have difficulty keeping drivers and warehouse workers, as exhibited by the recent labor strikes on Amazon. However, this handle can assist said workers with their jobs, making the job easier on their body, and reducing the risk of insurrection. As for students, the reasoning is fairly straightforward: if the teachers are happier and in less physical pain, then they are more likely to be kinder to the students. Yay!!